

Reading Support: Course Reflection

Companion to Week 13

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Admissibility has connected the course vocabulary

Supports: The same idea appeared under several names

Membership admits objects to sets. Functions admit one output for each domain value. Composition requires matching domains. Grammars license derivations. Machines accept configurations and strings. Models survive formal and empirical conditions.

The common question is which connections the abstraction permits and which it excludes.

MOLR Course synthesis

Descriptions and methods preserve different structures

Supports: Description and method must not be collapsed

A grammar describes a language through derivability. An automaton computes acceptance through configurations. CKY records licensed spans under a normal-form grammar. A transducer computes paired strings.

Equivalent stringsets do not erase these differences. The chosen method can add states, stacks, chart items, and intermediate categories absent from the linguistic claim.

MOLR Chs. 4–5

Equivalence always has a stated respect

Supports: Formal equivalence can conceal explanatory difference

Weakly equivalent grammars preserve terminal strings. Determinization preserves the accepted language. Regex identities preserve denotation. Normalization preserves membership under specified conditions.

None automatically preserves derivation, constituency, state interpretation, explanatory transparency, or acquisition consequences. “Equivalent” remains incomplete until the invariant is named.

MOLR Chs. 4–5

Ontogeny, continuity, and diversity test an explanation together

Supports: Three fundamental questions remain empirical constraints

A linguistic theory must allow children to construct a system from experience, explain the relation between developing and adult grammars, and admit the extensive variation found across languages.

A solution to one question that stipulates away either of the others relocates the problem. Contingent correspondences allow shared dependencies without requiring one predetermined realization.

MOLR §§2.3, 6.3

Grammar remains necessary within a wider activity

Supports: Grammar is indispensable without being sufficient

Successful linguistic action depends on grammatical organization together with meaning, context, knowledge, attention, social relations, and material opportunity. No one component ordinarily suffices.

An INUS analysis identifies grammar as a necessary part of one sufficient package while recognizing that other packages and repairs can sustain the activity.

MOLR §1.10

Contingent homomorphism makes the empirical claim explicit

Supports: Contingent homomorphisms preserve what languages actually sustain

A structure-preserving mapping becomes linguistic only when its source and target receive empirical interpretations and speakers sustain the relevant combinations. Another language may preserve the comparable relation with different categories or exponents.

The mathematics states what preservation amounts to. Acquisition, typology, and use determine which preservation claims belong to linguistic reality.

MOLR §6.2

Mathematics clarifies the claim and its route back to evidence

Supports: The destination is better linguistic questions

Mathematics contributes to linguistics by making abstractions, admissibility conditions, and consequences inspectable. It also shows where a result depends on a chosen domain, representation, or equivalence.

Its explanatory value lies in sharpening the empirical question: what does the formal relation preserve, across which variants, and what evidence could make us revise it?

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